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JEE MAIN PHYSICS DPP: FORCE, INERTIA & NEWTON'S 1ST LAW

Topic: Concept of Force, Types of Inertia, Equilibrium & Inertial Reference Frames | Time: 45 Min | Max Marks: 80

Marking Scheme: **+4, -1** | Designed & Curated by: **Syed Mazhar Ali**

Q1. Newton's first law of motion gives the qualitative definition of:

(a) Momentum
(b) Force and Inertia
(c) Acceleration
(d) Energy

Q2. The quantitative measure of inertia of a body in rectilinear motion is its:

(a) Velocity
(b) Volume
(c) Inertial mass
(d) Weight

Q3. A passenger in a moving bus is thrown forward when the bus suddenly stops. This phenomenon is an example of:

(a) Inertia of rest
(b) Inertia of direction
(c) Action-reaction pair
(d) Inertia of motion

Q4. A person sitting in an open train compartment travelling with uniform velocity tosses a ball vertically upwards. The ball falls:

(a) Exactly back into his hand
(b) Behind his hand
(c) Ahead of his hand
(d) To the left of his hand

Q5. In the previous scenario, if the train accelerates forward with constant acceleration a while the ball is in air, the ball lands:

(a) In his hand
(b) Behind his hand
(c) Ahead of his hand
(d) On the roof

Q6. When a wet dog shakes its body violently to dry its fur, water drops fly off tangent to the motion. This occurs primarily due to:

(a) Surface tension
(b) Gravitational force
(c) Inertia of direction and motion
(d) Centripetal force

Q7. A frame of reference attached to which of the following systems can be considered strictly non-inertial?

(a) A train moving at constant speed along a straight track
(b) A space capsule drifting in deep interstellar space with engines off
(c) An elevator descending with constant velocity
(d) A rotating carousel with constant angular velocity

Q8. A book of mass m rests on a flat table. The normal contact force exerted by the table on the book and the weight of the book:

(a) Are equal and opposite forces acting on the same body (Equilibrium pair)
(b) Form a Newton's third law action-reaction pair
(c) Act on different bodies
(d) Are gravitational forces

Q9. A particle of mass $m = 2$ kg is acted upon by three coplanar concurrent forces $\vec{F}_1 = (2\hat{i} + 3\hat{j})$ N, $\vec{F}_2 = (-5\hat{i} + \hat{j})$ N, and $\vec{F}_3$. If the particle remains in uniform motion with constant velocity $\vec{v} = (4\hat{i} - 2\hat{j})$ m/s, the force $\vec{F}_3$ must be:

(a) $(3\hat{i} + 4\hat{j})$ N
(b) $(3\hat{i} - 4\hat{j})$ N
(c) $(-3\hat{i} + 4\hat{j})$ N
(d) $(7\hat{i} - 2\hat{j})$ N

Q10. An astronaut in free interstellar space away from any gravitational influence is traveling at 1000 m/s. To keep moving at this constant speed in a straight line, the engine thrust required is:

(a) Proportional to his mass
(b) Dependent on fuel consumption
(c) Zero
(d) Inversely proportional to distance

Q11. A heavy block is suspended from a rigid ceiling by a string A, and an identical string B is attached to the bottom of the block. If a sudden sharp jerk is given to string B:

(a) String A breaks first
(b) Both strings break simultaneously
(c) Neither string breaks
(d) String B breaks first due to the inertia of rest of the block

Q12. In the previous question, if string B is pulled gradually and steadily with a continuously increasing force:

(a) String A breaks first because tension in A equals weight plus applied force
(b) String B breaks first
(c) Both break simultaneously
(d) The block accelerates upward

Q13. Dust is removed from a hanging carpet by beating it with a stick. The physical law underlying this technique is:

(a) Newton's second law
(b) Newton's first law (Inertia of rest)
(c) Newton's third law
(d) Law of conservation of linear momentum

Q14. A particle of mass m moves along a circular path of radius R at constant speed v. The net external force acting on the particle is:

(a) Zero, because speed is constant
(b) $\frac{mv^2}{R}$ directed tangent to the path
(c) $\frac{mv^2}{R}$ directed towards the center
(d) mg

Q15. A body is in translatory equilibrium when:

(a) Its kinetic energy is strictly zero
(b) Its speed is continuously increasing
(c) Its acceleration is directed towards a fixed point
(d) The vector sum of all external forces acting on it is zero ($\sum \vec{F}_{\text{ext}} = \vec{0}$)

Q16. When an automobile negotiates a sharp curve to the left, passengers experience an outward push towards the right. This perceived push is due to:

(a) Inertia of direction (tendency to continue in a straight line)
(b) Centripetal force exerted by seats
(c) A real outward magnetic force
(d) Gravitational attraction

Q17. Which of the following statements is **incorrect** regarding inertial frames of reference?

(a) Newton's first law is valid in all inertial frames
(b) A frame rotating with uniform angular velocity relative to an inertial frame is also inertial
(c) Any frame moving with constant velocity relative to an inertial frame is inertial
(d) A fictitious pseudo-force is not required in an inertial frame

Q18. A block of mass M is placed on a frictionless horizontal plane. A light string attached to the block passes over a frictionless pulley and carries a suspended mass m. The net force causing the acceleration of the entire combined system is:

(a) $(M+m)g$
(b) Mg
(c) mg
(d) $(M-m)g$

Q19. A bird is sitting in a large airtight glass cage held by a person. If the bird starts flying with uniform velocity inside the cage, the weight of the cage felt by the person:

(a) Increases
(b) Decreases
(c) Becomes zero
(d) Remains unchanged

Q20. Three concurrent forces $\vec{F}_1 = 6$ N (East), $\vec{F}_2 = 8$ N (North), and $\vec{F}_3$ keep a particle in static equilibrium. The magnitude and direction of $\vec{F}_3$ are:

(a) 10 N, $\tan^{-1}(4/3)$ West of South
(b) 10 N, $\tan^{-1}(3/4)$ East of North
(c) 14 N, 45° South of West
(d) 2 N, North-East

ANSWER KEY & STEP-BY-STEP SOLUTIONS

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Quick Answer Key

Q.No	Ans	Q.No	Ans	Q.No	Ans	Q.No	Ans
1	(b)	**6**	(c)	**11**	(d)	**16**	(a)
2	(c)	**7**	(d)	**12**	(a)	**17**	(b)
3	(d)	**8**	(a)	**13**	(b)	**18**	(c)
4	(a)	**9**	(b)	**14**	(c)	**19**	(d)
5	(b)	**10**	(c)	**15**	(d)	**20**	(a)

Detailed Solutions

Sol 1. Correct Option: (b)
Newton's first law states that an object continues in its state of rest or uniform motion in a straight line unless acted upon by an external net force. It defines the intrinsic property of **inertia** and qualitatively defines **force** as the agency that changes this state.

Sol 2. Correct Option: (c)
In translational mechanics, mass is the quantitative measure of inertia. A heavier mass requires a larger net force to achieve the same acceleration ($m = F/a$).

Sol 3. Correct Option: (d)
When the bus brakes suddenly, the lower part of the passenger's body comes to rest with the bus via friction, while the upper body tends to remain in motion due to **inertia of motion**, throwing the passenger forward.

Sol 4. Correct Option: (a)
In a frame moving with constant velocity $\vec{v}$, the ball shares the same horizontal velocity component $v_x = v$ as the train and the passenger throughout its vertical flight. With no horizontal acceleration ($a_x = 0$), it lands precisely back into the thrower's hand.

Sol 5. Correct Option: (b)
During the ball's time of flight T, the ball moves horizontally by $x_{\text{ball}} = v_0T$. The accelerated passenger travels $x_{\text{man}} = v_0T + \frac{1}{2}aT^2$. Since $x_{\text{man}} > x_{\text{ball}}$, the ball lands a distance $\frac{1}{2}aT^2$ behind his hand.

Sol 6. Correct Option: (c)
As the dog rapidly changes direction during shaking, the water droplets tend to maintain their instantaneous tangential straight-line velocity due to **inertia of direction** and detach when adhesive forces are overcome.

Sol 7. Correct Option: (d)
An inertial frame is non-accelerating. A rotating carousel has a non-zero centripetal acceleration $\vec{a} = -\omega^2\vec{r}$ directed towards the rotation axis, making it an intrinsically accelerated (non-inertial) reference frame.

Sol 8. Correct Option: (a)
Action-reaction pairs act on different interacting bodies (e.g., Earth on book vs. book on Earth). Here, the normal force $\vec{N}$ (table on book) and gravity $m\vec{g}$ (Earth on book) both act on the *same* body (the book) and sum to zero ($\vec{N} + m\vec{g} = \vec{0}$) to maintain static equilibrium.

Sol 9. Correct Option: (b)
Since the particle moves with constant velocity, its acceleration is zero ($\vec{a} = \vec{0}$). By Newton's first law:

$$\sum \vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \vec{0}$$

$$(2\hat{i} + 3\hat{j}) + (-5\hat{i} + \hat{j}) + \vec{F}_3 = \vec{0}$$

$$-3\hat{i} + 4\hat{j} + \vec{F}_3 = \vec{0} \implies \vec{F}_3 = 3\hat{i} - 4\hat{j}\ \text{N}$$

Sol 10. Correct Option: (c)
According to Galileo's law of inertia and Newton's first law, no net force is required to maintain motion at a constant velocity in a frictionless, field-free medium. Hence, required thrust = 0 N.

Sol 11. Correct Option: (d)
When a sudden sharp jerk is applied to bottom string B, the large inertia of rest of the heavy block prevents the force from instantaneously transmitting to upper string A. Hence, high tension develops instantaneously in B, causing string B to snap first.

Sol 12. Correct Option: (a)
When pulled slowly, the system is in quasi-static equilibrium. Tension in string A is $T_A = mg + F_{\text{pull}}$, while tension in string B is $T_B = F_{\text{pull}}$. Because $T_A > T_B$ at all times, string A reaches its breaking threshold first.

Sol 13. Correct Option: (b)
Beating the carpet sets the carpet fibers into rapid motion. The dust particles tend to stay in their state of rest due to **inertia of rest**, getting separated from the carpet and falling under gravity.

Sol 14. Correct Option: (c)
Although speed is constant, the direction of velocity changes continuously. This directional change constitutes an acceleration $\vec{a}_c = \frac{v^2}{R}$ directed towards the center. By Newton's laws, a net centripetal force $F_c = \frac{mv^2}{R}$ towards the center is mandatory.

Sol 15. Correct Option: (d)
Translational equilibrium requires that the linear acceleration of the center of mass is zero ($\vec{a}_{\text{cm}} = \vec{0}$), which is satisfied when the vector sum of all external forces is zero: $\sum \vec{F}_{\text{ext}} = \vec{0}$.

Sol 16. Correct Option: (a)
When the car steers left, the passenger's body tends to continue moving straight ahead along the original tangent due to **inertia of direction**. Relative to the left-turning car, this manifests as an outward lateral shift to the right.

Sol 17. Correct Option: (b)
Any reference frame rotating with angular velocity $\vec{\omega}$ experiences a centripetal acceleration ($\vec{a} = -\omega^2 \vec{r}$) and is therefore accelerated. Hence, rotating frames are non-inertial.

Sol 18. Correct Option: (c)
For the combined system of mass $(M + m)$, the normal force on M balances Mg, and internal string tensions cancel. The only unbalanced external driving force acting along the line of motion is the gravity acting on the hanging mass, which equals mg.

Sol 19. Correct Option: (d)
In an airtight sealed cage, the bird stays aloft by pushing air downward with a force equal to its weight. This downward air thrust is transmitted directly to the cage bottom, keeping the total measured weight equal to $(W_{\text{cage}} + W_{\text{bird}})$.

Sol 20. Correct Option: (a)
Resultant of $\vec{F}_1$ and $\vec{F}_2$:

$$\vec{F}_{12} = 6\hat{i} + 8\hat{j}$$

Magnitude $|\vec{F}_{12}| = \sqrt{6^2 + 8^2} = 10$ N.
For equilibrium, $\vec{F}_3 = -\vec{F}_{12} = -6\hat{i} - 8\hat{j}$.
Magnitude: $|\vec{F}_3| = 10$ N.
Direction: $\tan\theta = \frac{8}{6} = \frac{4}{3}$ West of South.